

# Technical specification

Valve-regulated lead-acid battery with Absorbent Glass Mat (AGM) separator. Sealed construction for standby power applications.

**12 V**

6 cells

**120 Ah**

10 h to 10.8 V at 25°C

**450 W/cell**

15 min to 1.67 V/cell at 25°C

## Electrical performance

|                       |                                  |
|-----------------------|----------------------------------|
| 10-hour capacity      | 120 Ah at 12.0 A to 10.8 V, 25°C |
| 20-hour capacity      | 126 Ah at 6.30 A to 10.8 V, 25°C |
| Internal resistance   | $\leq 4.0$ mOhms at 25°C         |
| Maximum discharge     | 840A for 5 seconds at 25°C       |
| Self-discharge        | 3% average per month at 20°C     |
| Design life / standby | 12 years                         |

## Charging and operating conditions

|                          |                                                            |
|--------------------------|------------------------------------------------------------|
| Charge method            | Constant voltage                                           |
| Float / cycle voltage    | 13.38-13.80 V / 14.40-14.70 V at 25°C                      |
| Maximum charge current   | 37.8 A                                                     |
| Temperature compensation | Float: -20 mV/°C per battery; cycle: -30 mV/°C per battery |
| Operating temperatures   | Discharge: -20 to +60°C; charge: -10 to +60°C              |
| Storage temperature      | -20 to +60°C                                               |

## Mechanical specification

|                        |                                           |
|------------------------|-------------------------------------------|
| Dimensions (L × W × H) | 410 × 176 × 224 mm; overall height 224 mm |
| Approximate weight     | 36.5 kg $\pm 5\%$                         |
| Terminal / torque      | See terminal drawing; Not specified       |
| Container              | ABS, UL94-HB                              |

# Construction & applications

High-output AGM construction for UPS standby power. Computer-designed grids, optimized paste and lead-calcium-tin plates support high energy density. Design life: 12+ years at 20°C.

## Battery construction

|                 |                   |
|-----------------|-------------------|
| Positive plate  | Lead dioxide      |
| Negative plate  | Lead              |
| Container / lid | ABS / ABS         |
| Safety valve    | Rubber            |
| Terminals       | Copper            |
| Separator       | Glass fibre (AGM) |
| Electrolyte     | Sulfuric acid     |

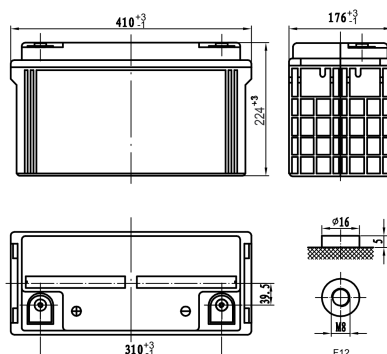
## Features

- Insulated intercell connections.
- Gas recombination efficiency up to 99%.
- High-rate discharge performance.
- Special welding process.
- Self-regulating valves; no water topping-up.
- Shelf life without recharge: 6 months at 20°C.
- Low internal pressure; special AGM separators.
- Optimized for 5–15 min; loads up to 1 hour.

## Applications

- Uninterruptible power supplies (UPS).
- Telecommunications backup power.
- Power and energy infrastructure.
- Security and fire alarm systems.
- Renewable-energy systems.
- Medical equipment.
- Emergency lighting.
- Data centres.

## Dimensions and terminal details / mm



Dimensions and tolerances govern; do not scale the drawing.

# Discharge performance

Fully charged battery / 25°C

## Constant-current discharge / A

| V/cell | 3m     | 5m     | 10m    | 15m    | 20m    | 25m    | 30m    | 45m    | 1h    |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|-------|
| 1.60   | 487.00 | 414.00 | 340.00 | 266.00 | 223.00 | 189.00 | 160.00 | 111.00 | 86.10 |
| 1.65   | 471.00 | 405.00 | 318.00 | 255.00 | 211.00 | 179.00 | 151.00 | 105.00 | 81.60 |
| 1.67   | 465.00 | 402.00 | 309.00 | 250.00 | 206.00 | 175.00 | 147.00 | 103.00 | 79.80 |
| 1.70   | 446.00 | 384.00 | 294.00 | 244.00 | 198.00 | 168.00 | 143.00 | 99.80  | 77.70 |
| 1.75   | 427.00 | 368.00 | 287.00 | 234.00 | 191.00 | 161.00 | 135.00 | 94.50  | 74.60 |
| 1.80   | 406.00 | 362.00 | 282.00 | 223.00 | 183.00 | 157.00 | 131.00 | 92.40  | 72.50 |

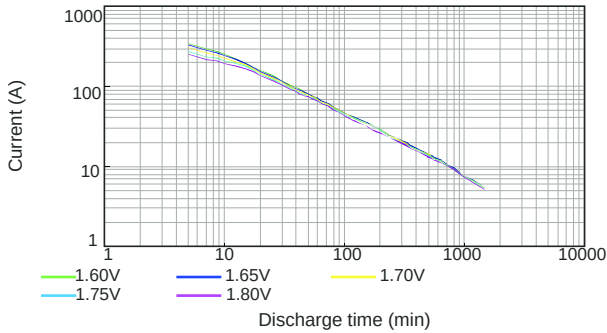
## Constant-power discharge / W per cell

| V/cell | 3m     | 5m     | 10m    | 15m    | 20m    | 25m    | 30m    | 45m    | 1h     |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 1.60   | 904.00 | 764.00 | 593.00 | 475.00 | 403.00 | 341.00 | 294.00 | 203.00 | 161.00 |
| 1.65   | 886.00 | 741.00 | 575.00 | 457.00 | 386.00 | 324.00 | 278.00 | 197.00 | 156.00 |
| 1.67   | 879.00 | 732.00 | 568.00 | 450.00 | 379.00 | 317.00 | 271.00 | 194.00 | 154.00 |
| 1.70   | 856.00 | 699.00 | 548.00 | 440.00 | 371.00 | 311.00 | 269.00 | 191.00 | 151.00 |
| 1.75   | 832.00 | 667.00 | 538.00 | 424.00 | 358.00 | 301.00 | 261.00 | 185.00 | 147.00 |
| 1.80   | 813.00 | 643.00 | 510.00 | 406.00 | 338.00 | 287.00 | 249.00 | 180.00 | 143.00 |

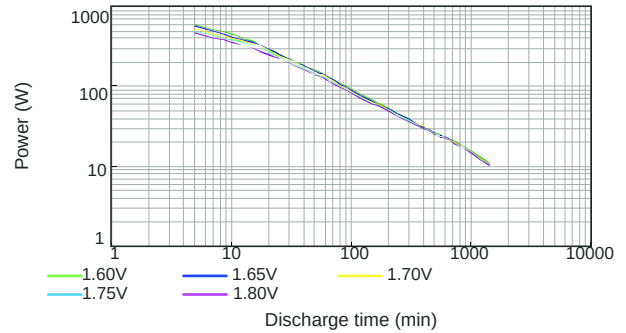
End voltage is per cell. Multiply W/cell by 6 for power per 12 V battery. Current is not divided by 6. Values are averages obtained within three charge/discharge cycles, not guaranteed minimums. Fully charge before testing. Product information may change without notice.

### Characteristic curves

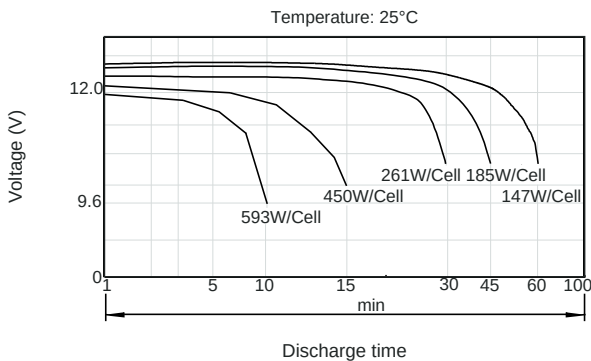
#### Constant-current discharge



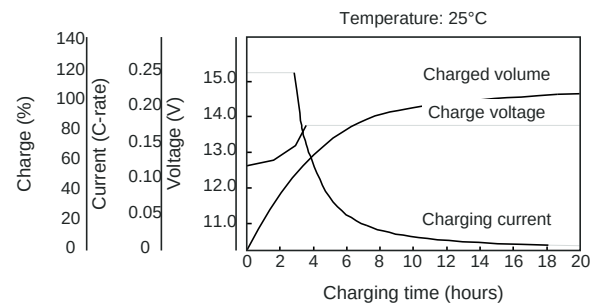
#### Constant-power discharge



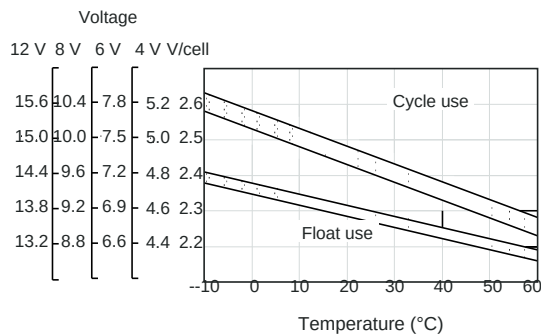
#### Discharge characteristics



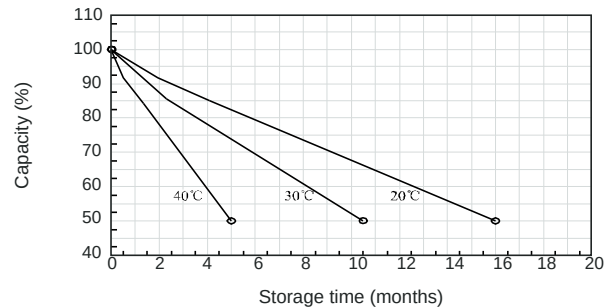
#### Charging characteristics



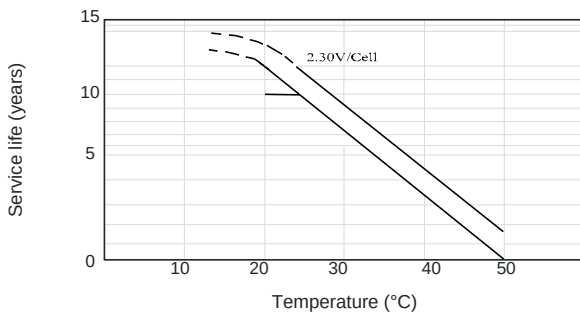
#### Charge voltage vs. temperature



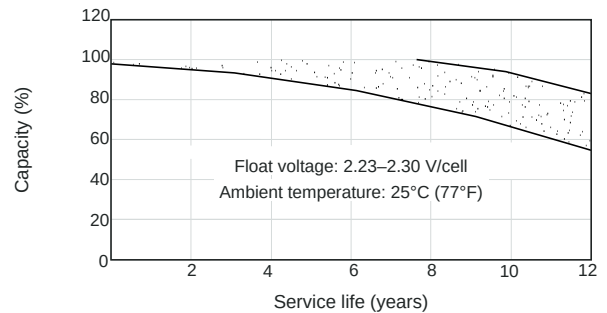
#### Self-discharge



#### Float life vs. temperature



#### Standby service life



Typical characteristic curves. Model-specific ratings are given on pages 1 and 3. Design life is not a warranty.